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A PROJECT REPORT ENTITLED

SPATIAL HOTSPOTS OF MPOX RISKS IN THE DEMOCRATIC REPUBLIC OF CONGO: A HEALTH-ZONE ANALYSIS

BY

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INTRODUCTION

1.1 Statement of the Problem

Monkeypox is a newly developing zoonotic infection caused by monkeypox virus (MPXV) which has an incubation period of up to 21 days. The MPXV is a large double-stranded DNA virus belonging to the Ortho poxvirus genus. Monkeypox viruses are genetically of two distinct phylogenetic clades, one is the Central African clade, the other is West African clade (Zhang et al., 2022). In endemic areas, various mammalian species can carry MPXV, and periodically the transmission of the virus to human will take place (Petersen et al., 2019). There are three potential routes of exposure to the virus, interaction with wild animals, close proximity to sick individuals, and contact with infectious fomites.

Monkeypox was first reported by Preben von Magnus in 1958 in laboratory cynomolgus monkeys. The first human monkeypox virus infection was reported in Democratic Republic of Congo (DRC) in 1970). After that, MPXV infection was discovered in 11 African countries. Once considered a rare, sporadic infection in humans, recent studies in the Democratic Republic of the Congo (DRC, previously Zaire), where most cases have been reported, suggest that the incidence of human MPX has markedly increased in the Congo Basin since the 1980s (McCarthy, 2022).

1.1.1 Mpox Situation in DRC

In 2024, as of 26 May, a total of 7 851 Mpox cases were reported in the Democratic Republic of Congo, including 384 deaths (CFR 4.9%). These cases were reported in 177 of the 519 (34%) health zones across 22 out of the 26 provinces (85%). The most affected provinces in 2024 are Equateur, Sud Ubangi, Sankuru and South Kivu. The number of cases reported weekly remains consistently high while the outbreak continues to expand geographically. High test positivity among tested cases in most provinces also suggests that undetected transmission is likely ongoing in the community (WHO, 2025). Figure 1 below shows the Geographic distribution of Mpox cases from March to May by health zone in the Democratic Republic of Congo.

B.-A. M. Mandja et al., conducted spatial and temporal clusters analysis of weekly monkeypox cases collected from 2000 to 2015 at the health zone scale in 2019. They reported peaks of cases occurred from July to September for the 2000–2002 and 2003–2009 sub-periods and from January to March for the 2010–2015 sub-period. Despite the lack of additional data for confirmation, the increasing of monkeypox reported incidence was observed in the Democratic Republic of Congo during 2000–2015 period. Figure 2 shows the 2010–2015 sub-period Mpox risk analysis of the study conducted.

Understanding spatial risk is essential for targeted surveillance and resource allocation. Therefore, this project seeks to map and evaluate spatial concentration of weekly Mpox cases across 200 health zones in Democratic Republic of Congo (DRC) from October 2025 to December, 2025

1.2 Project Objectives

The objectives of this project are to;

i. compute weekly based Mpox risk at health zone level and produce a risk map.
ii. identify spatial concentration of the Mpox risk across the health zones
iii. compare clustering under contiguity-based and distance-based spatial weight
iv. provide insight for resource allocation.

1.3 Methods to be Used

i. Queen Contiguity
ii. Fixed Distance
iii. k-Nearest Neighbors

1.4 Data

i. DRC weekly Mpox data from World Health Organization (WHO)
ii. DRC health zone boundaries (shapefile) from GADM
iii. Population at Risk: World Pop population counts from World Pop

1.5 Analysis

i. Data preparation
ii. Zonal Statistics to obtain population at risk
iii. Weekly risk indicator
iv. Spatial weights (contiguity based and distance based)
v. Clustering Assessment

1.6 Expected Outcomes

i. A choropleth map of weekly Mpox risk in DRC from October to December, 2025
ii. Localized hotspot maps of Mpox risk at health zone level
iii. Comparison of contiguity based and distance-based methods used
iv. Recommendations for resource allocation at hotspots health zone

1.7 Organization of Report

This project is divided into six (6) chapters. Chapter 1 is the introductory chapter which gives a general overview of the project carried out. Chapter 2 states the relevant information about the study area. A review of relevant literature is provided in Chapter 3. Chapter 4 addresses both the materials and methods used for the project. Chapter 5 consists of results, comparison and discussions and Resource allocation and conclusion is Chapter 6

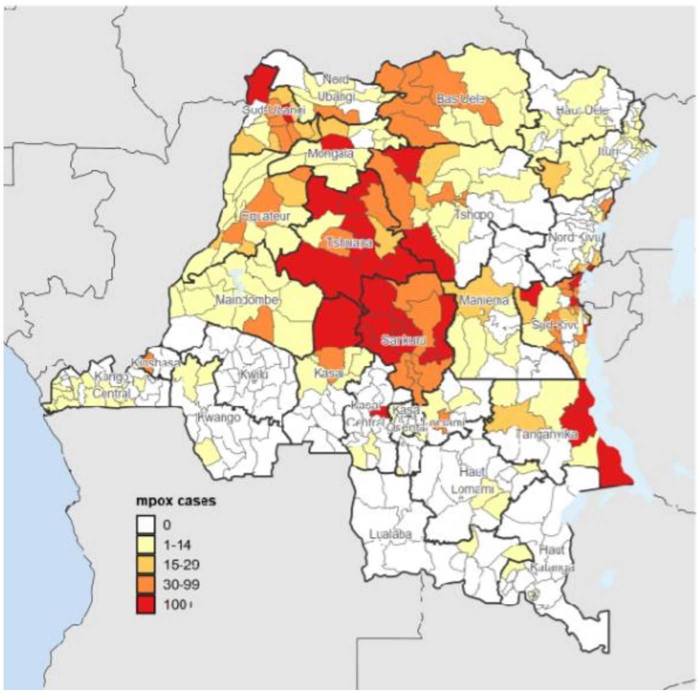


Figure 1: Geographic distribution of Mpox cases from March to May by health zone in the Democratic Republic of the Congo

***Source**: DRC Ministry of Public Health, (2025)*

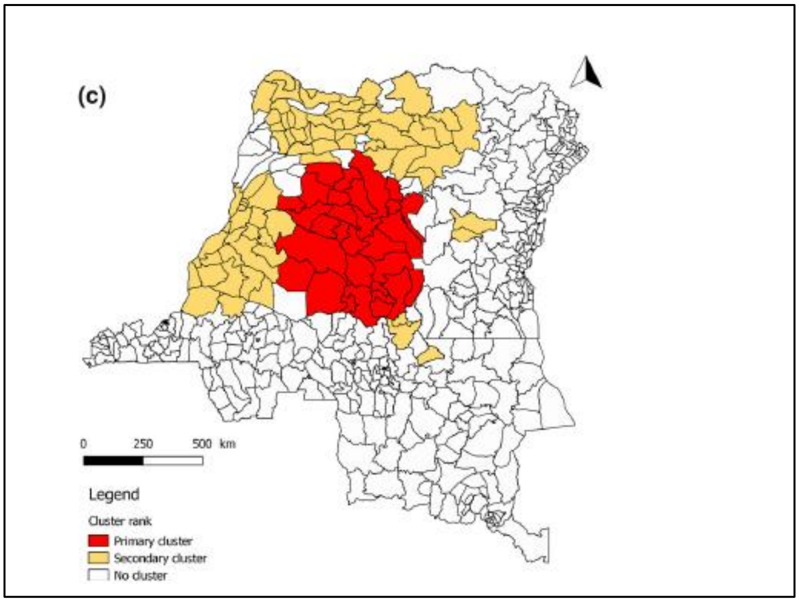


Figure 2: A purely spatial scan analysis of monkeypox relative risk infection in the Democratic Republic of Congo, 2010–2015

Source: *B.-A. M. Mandja et al., (2019)*

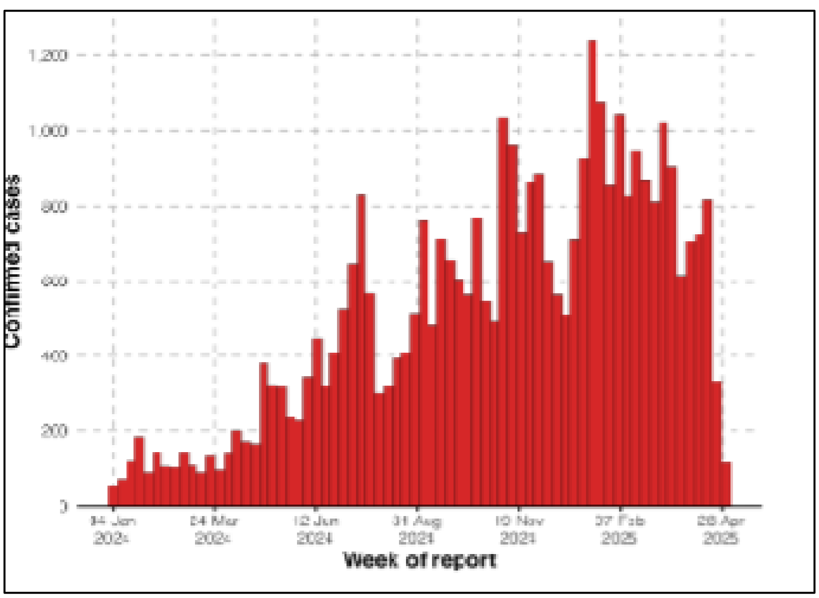


Figure 3: Epidemic curves of confirmed Mpox cases reported in the DRC,1 January 2024 – 11 May 2025

Source: *WHO (20*

CHAPTER 2

STUDY AREA

2.1 Brief History of Democratic Republic of Congo (DR Congo)

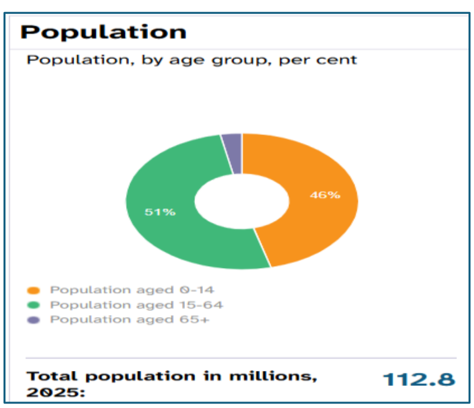
The territory that is now the Democratic Republic of the Congo has been inhabited for at least 90,000 years. Significant population and technological transformations occurred during the first millennium BC with the spread of Bantu-speaking peoples. This expansion led to the formation of organized political entities, including early confederations in the Pool Malebo area and later major states such as the Kingdom of Kongo and the Luba and Lunda Empires (Cordell, 2026).

2.2 Location, Size and Geography

The Democratic Republic of the Congo is located in central sub-Saharan Africa and borders nine countries, as well as the South Atlantic Ocean. It lies between 6° N and 14° S and 12° E and 32° E, straddling the Equator with most of its territory in the Southern Hemisphere. With an area of about 2.35 million km², it is the second-largest country in Africa after Algeria **(Anon, 2021)**

2.3 Population

As of 2025, the population of the Democratic Republic of the Congo (DRC) is estimated at approximately 112–114 million people, making it one of the most populous countries in Africa. The country is now divided into 26 provinces, including the city-province of Kinshasa as the capital; these provinces were established through administrative decentralization to replace the former ten-province structure (Anon., 2025).



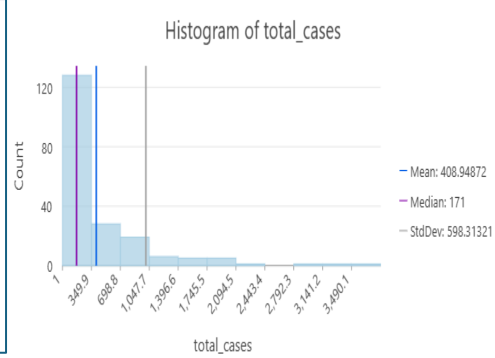


Figure 4 Histogram of Mpox Data Used

Figure 5: Population of DR Congo, 2025

***Source**: United Nations Population Fund (2025)*

CHAPTER 3

LITERATURE REVIEW

3.1 Spatial Clustering

Spatial clustering can be defined as the process of grouping objects with certain dimensions into groups such that objects within a group exhibit similar characteristics when compared to those which are in the other groups. It is an important part of spatial data since it provides certain insights into the distribution of data and characteristics of spatial clusters (Neethu and Surendran, 2013). Cluster detection methods are increasingly being used in public health surveillance efforts. As data is updated, health departments can use spatial and space-time clustering methods to search for unusual patterns and the detection of disease outbreaks ((Rushton 2003); (Mandl et al. 2004). Knowing whether or not a cluster exists and where they are located is useful for making decisions and formulating health research and policies (Blanford, J.I. 2021)

3.2 Spatial Autocorrelation

Spatial autocorrelation measures the direction of the linear association between the variables and the degree of intensity of the spatial pattern of a given variable with the same variable, but for a defined neighborhood (Dubé and Legros, 2014). Spatial autocorrelation quantifies the relationship between attribute values at one location and those at neighboring locations, with positive spatial autocorrelation indicating similarity among nearby values and negative spatial autocorrelation indicating dissimilarity. (Diniz-Filho et al. 2012).

3.3 Global Measure of Spatial Autocorrelation

The aim of global cluster analysis is to determine the presence or absence of clustering in the whole study region. There are numerous methods for testing global existence of clusters. The most widely used measure of global clustering in epidemiology is the method proposed by Moran (1950). Moran's Index is a weighted correlation coefficient that is used to measure deviation from spatial randomness (Pfeiffer et al. 2008).

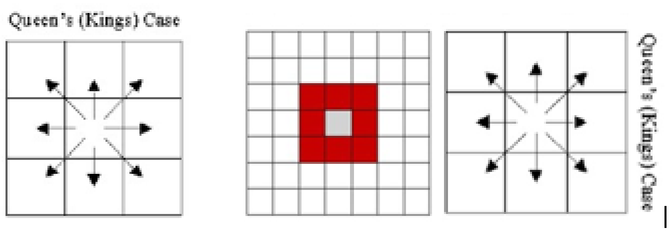
3.4 Local Measure of Spatial Autocorrelation (Anselin's local Moran's I)

Anselin (1995) introduced Local Indicators of Spatial Association (LISA) as local statistics that decompose global measures of spatial autocorrelation, enabling the identification of local clusters and spatial outliers.

3.5 Distance-Based Spatial Weight

3.5.1 Queen Contiguity

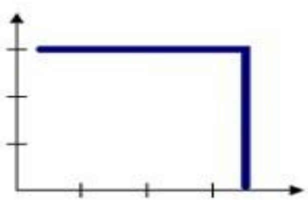
The Queen Contiguity is a Spatial neighbors based on Adjacency or neighborhood (that whether or not areas share a common boundary) considers a neighborhood of all eight cells. All cells that touch the central cell (adjacent and diagonal cells) are considered.



Source: *Blanford, J.I. (2021)*

3.5.2 Fixed Distance Band

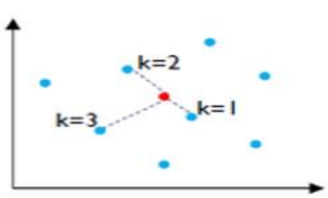
For Fixed Distance Band, everything within a specified distance of each feature is included in the analysis and everything outside is excluded. This is set as the distance threshold. The method is based on Euclidean distance.



Source: Table 5.3 (Blanford, 2025, p. 11)

3.5.3 k-Nearest Neighbors

The closest k features are included in the analysis. Each feature is assessed within the spatial context of a specified number of its closest neighbors. If K (the number of neighbors) is 8, then the eight closest neighbors to the target features will be included in the computations for that feature. Neighbors are based on Euclidean distance.



*Source: **Table 5.3** (Blanford, 2025, p. 11)*

CHAPTER 4

MATERIALS AND METHODS

4.1 Materials Used

Secondary weekly reported Mpox surveillance data obtained from World Health Organization (WHO), aggregated at the health-zone level in the Democratic Republic of the Congo were employed in this project. The format of the data was in CSV file. The software used to achieve the objective of this project was ArcGIS Pro.

4.1.1 Data Source

This project employed 200 Secondary weekly reported Mpox surveillance data from October to December 2025 (3 months period) obtained from World Health Organization (WHO) and aggregated at the health-zone level. Health-zone boundary (shapefile) with 519 health zones was downloaded from GADM and Population at risk was obtained from WorldPop population raster 2025 with 100m resolution. The coordinate system used was Africa Albers Equal Area Conic.

	A	B	C	D	E	F	G	H
1	province	health_zone	case_total	death_total	cfr	case_recent	death_recent	last_reported
2	Sud-Kivu	Miti-Murrhesa	8975	11	0.001225627	627	0	12/7/2025
3	Sud-Kivu	Nyangezi	4823	2	4.15E-04	401	0	12/7/2025
4	Nord Kivu	Karisimbi	4546	0	0	380	0	12/7/2025
5	Sankuru	Djalo-Ndjeka	895	1	0.001117318	274	0	12/7/2025
6	Sankuru	Omendjadi	966	2	0.002070393	262	0	12/7/2025
7	Sud-Kivu	Nyantende	2202	1	4.54E-04	207	0	12/7/2025
8	Tshuapa	Lingomo	3312	158	0.047705314	197	12	12/7/2025
9	Tshuapa	Ikela	3230	22	0.006811146	193	5	12/7/2025
10	Sankuru	Lodja	1283	6	0.004676539	180	0	12/7/2025
11	Sankuru	Tshumbe	1712	5	0.002920561	178	0	12/7/2025
12	Sankuru	Lusambo	1673	17	0.010161387	170	0	12/7/2025
13	Sankuru	Tshudi-Loto	1413	26	0.018400566	151	0	12/7/2025
14	Sud-Kivu	Minova	1670	2	0.001197605	150	0	12/7/2025
15	Sankuru	Dikungu	2497	53	0.021225471	146	0	12/7/2025
16	Sud-Kivu	Katana	909	1	0.00110011	145	0	12/7/2025
17	Kasai	Mushenge	1073	28	0.026095061	141	5	12/7/2025
18	Sud-Kivu	Kabare	975	0	0	139	0	12/7/2025
19	Sankuru	Bena-Dibele	1341	1	7.46E-04	127	0	12/7/2025
20	Sankuru	Lomela	1343	50	0.037230082	118	0	12/7/2025

Figure 6: A sample of the Mpox Data Used

4.1.2 Data Preparation and Cleaning

The Mpox CSV data obtained from WHO was joined to the DRC health-zone polygons using a joined key. Population at risk was obtained by aggregating the WorldPop population raster to health zones using zonal statistics (SUM) and joining the resulting population totals to the polygons. The **total_cases** field was then calculated directly in the polygon attribute table to ensure that each health zone contained both the numerator (case_total) and denominator (population at risk). Missing data were recorded as NULL. The full national layer was retained for presentation mapping with “No data” zones clearly symbolized.

4.2 Methods Used

Autocorrelation and Mapping of Clusters

4.2.1 Global Spatial Autocorrelation (Global Moran's I)

Queen Contiguity

Geoprocessing→ Spatial Statistics Tools →Analyzing Pattern → Spatial Autocorrelation (Global Moran's I)

i. Mpox_Data_Projected was used as the Input Feature
ii. total_cases field was used as the input field
iii. CONTIGUITY_EDGES_CORNERS (Queen) was chosen as the Conceptualization of Spatial Relationships
iv. Row Standardization was checked
v. The analysis was run and recorded Moran's I, z-score, p-value

Fixed Distance Band

i. Mpox_Data_Projected was used as the Input Feature
ii. total_cases field was used as the input field
iii. Fixed Distance Band was chosen as the Conceptualization of Spatial Relationships
iv. Row Standardization was checked
v. Default distance threshod and 100000 threshold was set
vi. The analysis was run and recorded Moran's I, z-score, p-value

k-Nearest Neighbors

i. Mpox_Data_Projected was used as the Input Feature
ii. total_cases field was used as the input field
iii. K_NEAREST_NEIGHBORS was chosen as the Conceptualization of Spatial Relationships
iv. Number of Neighbors were set to 4 and 8
v. The analysis was run and recorded Moran's I, z-score, p-value

4.2.2 Local Clustering and Outliers Analysis (Anselin Local Moran's I)

Geoprocessing→ Spatial Statistics Tools →Mapping Clusters → Cluster and Outlier Analysis (Anselin Local Moran's I)

Steps (Repeated for Queen, kNN, Fixed Distance)

i. Input Features: **Mpox_Data_Projected**

ii. Input Field: **total_cases**

Queen Contiguity

Conceptualization of Spatial Relationships**: CONTIGUITY_EDGES_CORNERS (Queen)**

Fixed Distance Band

Conceptualization of Spatial Relationships: **FIXED_DISTANCE_BAND**

Distance threshold: **Default** and **100 Km**

k-Nearest Neighbors

Conceptualization of Spatial Relationships: **k-Nearest Neighbors**

Number of Neighbors were set to **4** and **8**

iii. Run → output layer includes:

a. Cluster type map (HH, LL, HL, LH)

b. Moran's scatter plot

c. Histogram plot of the input field

CHAPTER 5

RESULTS, COMPARSION AND DISCUSSION

5.1 Results

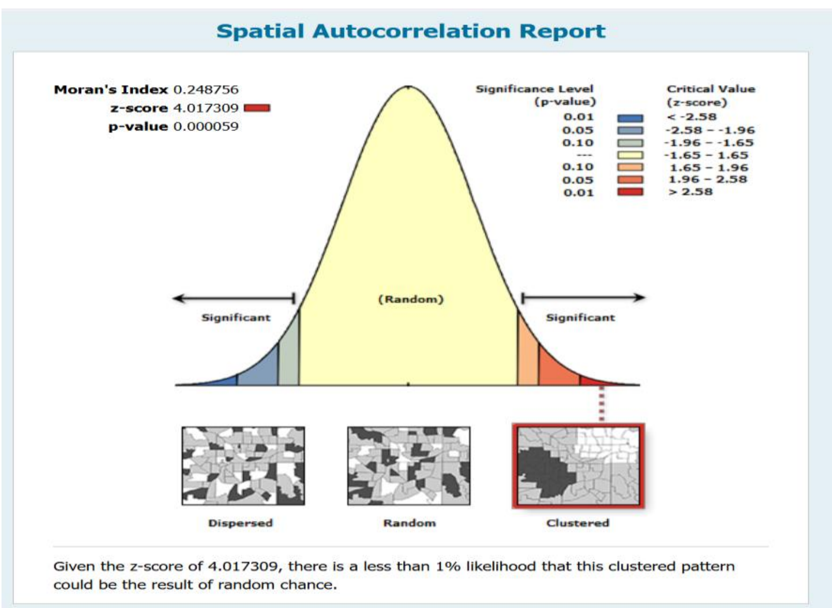
The primary objective of this project was to use Contiguity Weight and Distance Weight spatial neighborhoods structures to detect autocorrelation and clustering (Hotspots) of Mpox disease outbreaks in 200 health zones in Democratic Republic of Congo.

5.1.1 Global Spatial Autocorrelation (Global Moran's I)

Global spatial autocorrelation of Mpox cases across health zones in the Democratic Republic of the Congo was assessed using Global Moran's I under 3 spatial neighborhoods structures. The results for each spatial weight are shown below.

Queen Contiguity

Using Queen contiguity (edges and corners), Mpox cases exhibited significant positive spatial autocorrelation (Moran's I = 0.249, z = 4.02, p < 0.001), indicating that adjacent health zones tended to report similar numbers of Mpox cases.



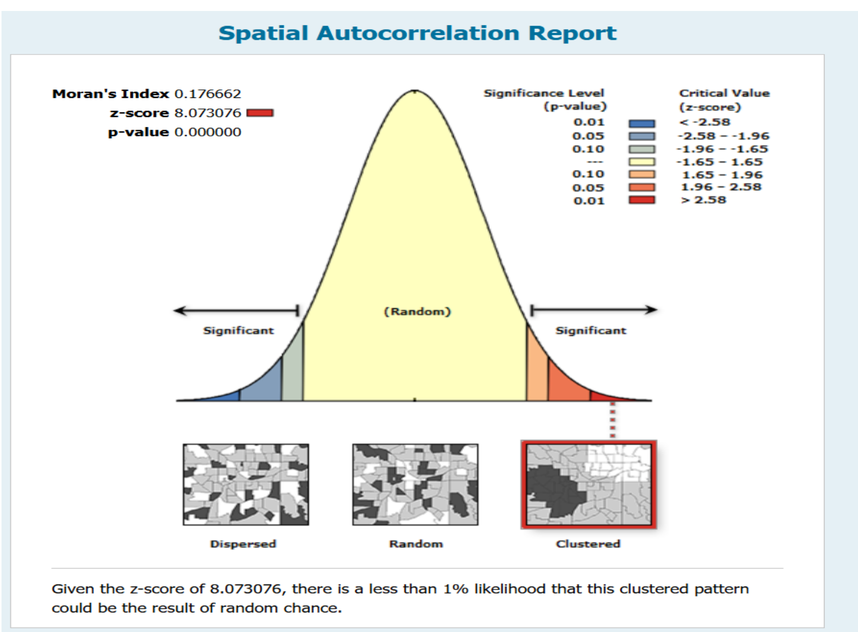
Global Moran's I Summary	
Moran's Index	0.248756
Expected Index	-0.005155
Variance	0.003995
z-score	4.017309
p-value	0.000059

Dataset Information	
Input Feature Class:	Mpox_Data_Projected
Input Field:	TOTAL_CASES
Conceptualization:	CONTIGUITY_EDGES_CORNERS
Distance Method:	EUCLIDEAN
Row Standardization:	True
Distance Threshold:	None
Weights Matrix File:	None

Figure 7: Reports for Spatial Autocorrelation of Queen Contiguity

Fixed Distance Band

When spatial relationships were defined using fixed distance bands, Global Moran's I achieved positive spatial autocorrelation at both the default distance threshold (~328 km; Moran's I = 0.177, z = 8.07, p < 0.001) and the more restrictive 100 km threshold (Moran's I = 0.231, z = 3.92, p < 0.001). The difference in Moran's I magnitude across distance thresholds indicates that the strength of global spatial autocorrelation varies with the spatial scale used to define neighborhood interactions, while the results showed that global spatial dependence in Mpox cases is present across both shorter and broader regional distance scales.

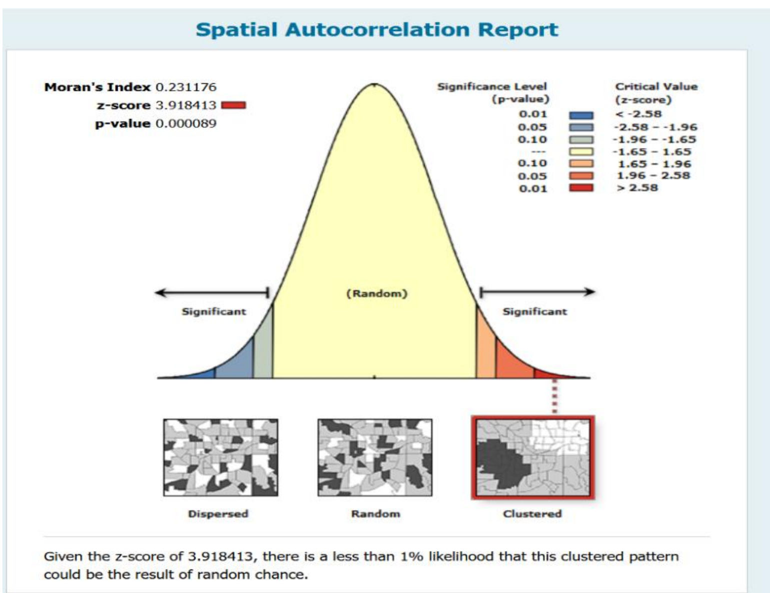


Global Moran's I Summary	
Moran's Index	0.176662
Expected Index	-0.005155
Variance	0.000507
z-score	8.073076
p-value	0.000000

Dataset Information	
Input Feature Class:	Mpox_Data_Projected
Input Field:	TOTAL_CASES
Conceptualization:	FIXED_DISTANCE
Distance Method:	EUCLIDEAN
Row Standardization:	True
Distance Threshold:	328403.0261 meters
Weights Matrix File:	None
Selection Set:	False

Figure 8: Reports for Spatial Autocorrelation of Fixed Distance (Default) Threshold

Fixed Distance Band (100 km)



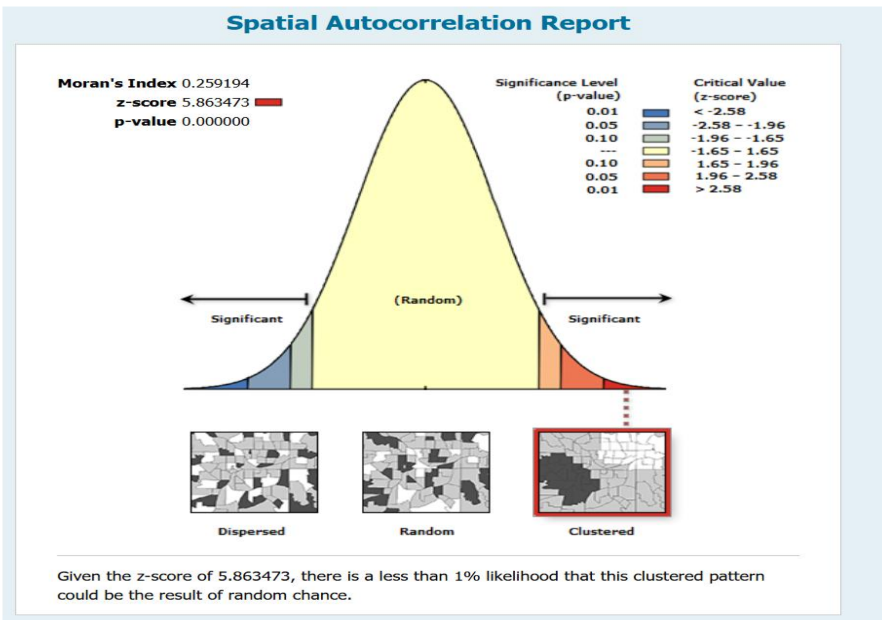
Global Moran's I Summary	
Moran's Index	0.231176
Expected Index	-0.005155
Variance	0.003638
z-score	3.918413
p-value	0.000089

Dataset Information	
Input Feature Class:	Mpox_Data_Projected
Input Field:	TOTAL_CASES
Conceptualization:	FIXED_DISTANCE
Distance Method:	EUCLIDEAN
Row Standardization:	True
Distance Threshold:	100000.0000 meters
Weights Matrix File:	None
Selection Set:	False

Figure 9: Reports for Spatial Autocorrelation of Fixed Distance for 100 km threshold

k-Nearest Neighbors (4 Neighbors)

For k-nearest neighbor criteria, global spatial autocorrelation was consistent across neighborhood sizes. The k = 4 neighbors had the greatest Global Moran's I value (0.259, z = 5.86, p < 0.001). This suggests a strong global spatial dependence. At k = 8, the Global Moran's I remained significantly positive (I = 0.232, z = 7.50, p < 0.001). Although the Moran's I value is slightly lower than at k = 4, the higher z-score provides greater evidence against spatial randomness in the k = 8 neighborhood. With k = 8, Global Moran's I remained significantly positive (I = 0.232, z = 7.50, p < 0.001).



Global Moran's I Summary

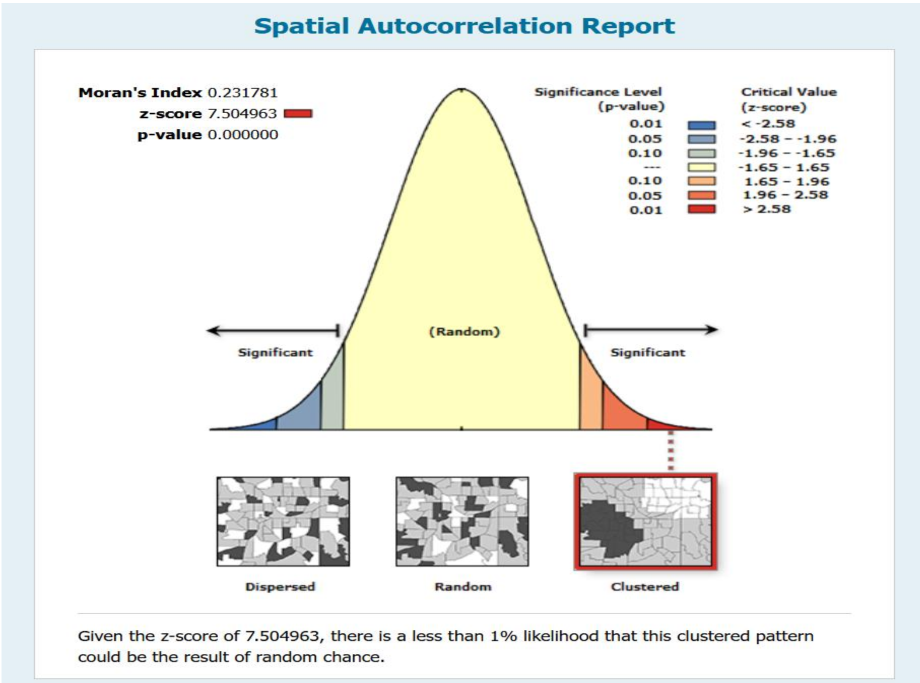
Moran's Index	0.259194
Expected Index	-0.005155
Variance	0.002033
z-score	5.863473
p-value	0.000000

Dataset Information

Input Feature Class:	Mpox_Data_Projected
Input Field:	TOTAL_CASES
Conceptualization:	K_NEAREST_NEIGHBORS
Distance Method:	EUCLIDEAN
Row Standardization:	True
Distance Threshold:	None
Weights Matrix File:	None
Selection Set:	False

Figure 10: Reports for Spatial Autocorrelation of k-nearest neighbors of 4

k-Nearest Neighbors (8 Neighbors)



Global Moran's I Summary

Moran's Index	0.231781
Expected Index	-0.005155
Variance	0.000997
z-score	7.504963
p-value	0.000000

Dataset Information

Input Feature Class:	Mpox_Data_Projected
Input Field:	TOTAL_CASES
Conceptualization:	K_NEAREST_NEIGHBORS
Distance Method:	EUCLIDEAN
Row Standardization:	True
Distance Threshold:	None
Weights Matrix File:	None
Selection Set:	False

Figure 11: Reports for Spatial Autocorrelation of k-nearest neighbors of 8

In conclusion, regardless of whether neighborhoods are defined by adjacency (Queen), distance bands (default and 100 km), or k-nearest neighbors (k = 4, 8), the results consistently reject spatial randomness and support the conclusion that Mpox cases show significant positive global spatial autocorrelation across DRC health zones, with the estimated strength of global autocorrelation varying by neighborhood.

5.1.2 Local Clustering and Outlier Analysis (Anselin Local Moran's I)

For the local Clustering and Outlier Analysis, each map shows statistically significant local spatial associations of Mpox cases at the health-zone level:

i. High–High (**HH**) means Health Zones with high Mpox counts surrounded by high-count neighbors. This indicates **hotspots areas**

ii. Low–Low (**LL**) means health zones with low Mpox counts surrounded by low-count neighbors. This indicates **cold spots areas**

iii. High–Low (**HL**) means high-value zones surrounded by low values, indicating **spatial outliers**

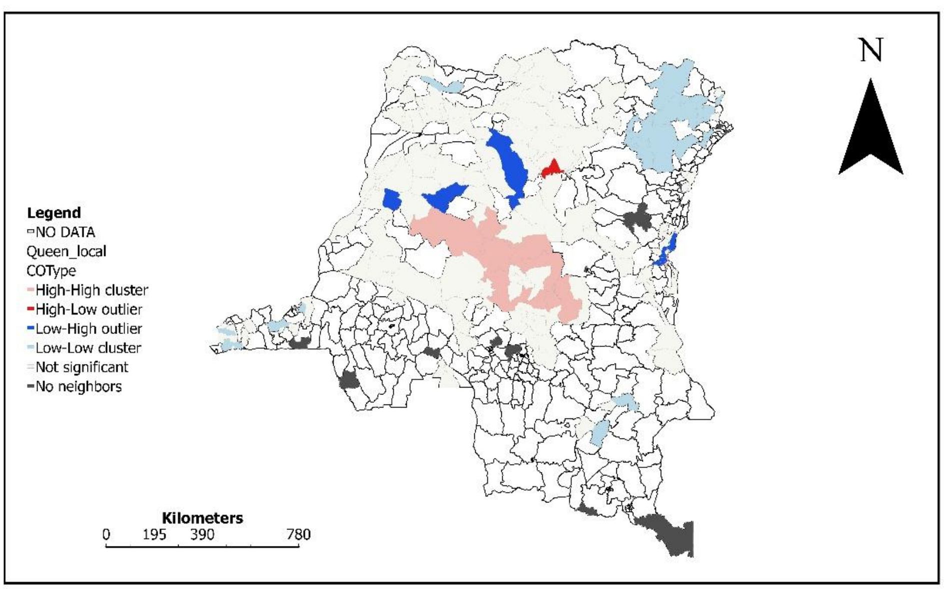
iv. Low–High (**LH**): low-value zones surrounded by high values → **spatial outliers**

v. Not significant: no statistically significant local pattern.

Queen Contiguity

The Local Moran's I map based on queen contiguity reveals a prominent High–High cluster concentrated in the central part of DR Congo. This shows spatially contiguous health zones with high Mpox case counts. In contrast, Low–Low clusters dominate the northeastern part of the country, suggesting spatially stable areas of lower Mpox burden. A limited number of spatial outliers are observed at the margins of the central cluster and in isolated southern locations, indicating localized deviations from surrounding patterns.

Mapping of Clusters by Queen Contiguity



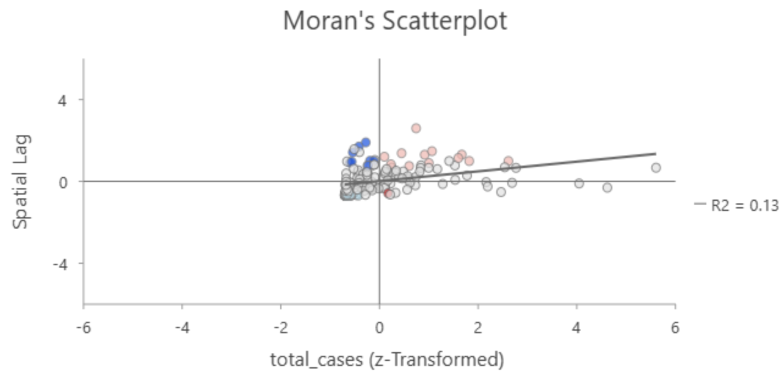
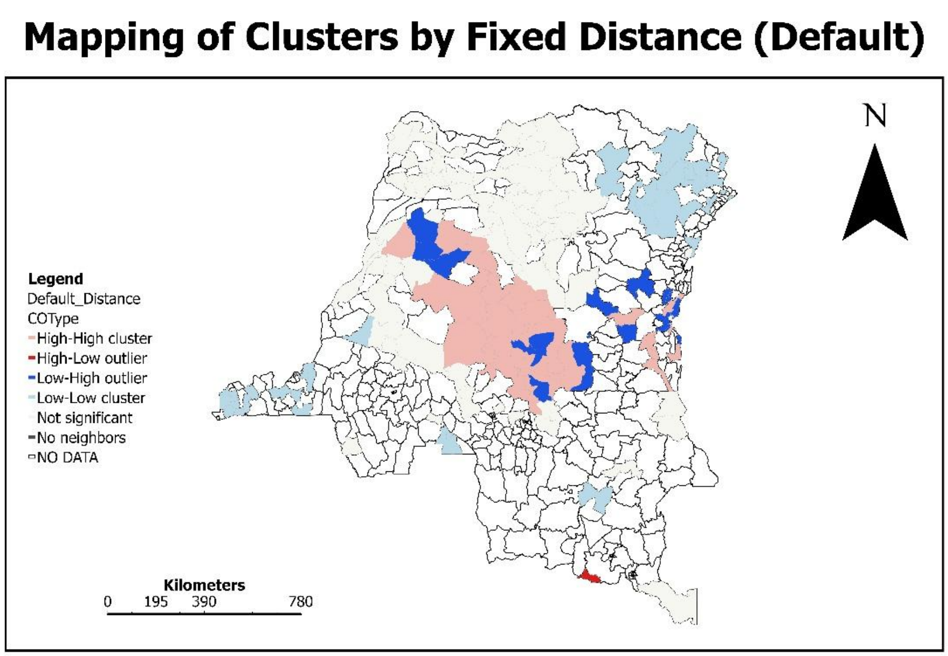


Figure 12: Result for local cluster by Queen Contiguity

Fixed Distance Band (Default)

Under the default fixed distance weighting, High–High Mpox cluster became more extensive in the health zones located in the central and extreme eastern, covering a broader contiguous area compared to the contiguity-based results. The method also revealed an increase in number of spatial outliers, particularly along the eastern, northwestern and some parts of the margins of the central cluster, indicating zones with Mpox levels that contrast with their surrounding neighbors. Low–Low clusters remain concentrated in the northeastern and southern part of the country**.**



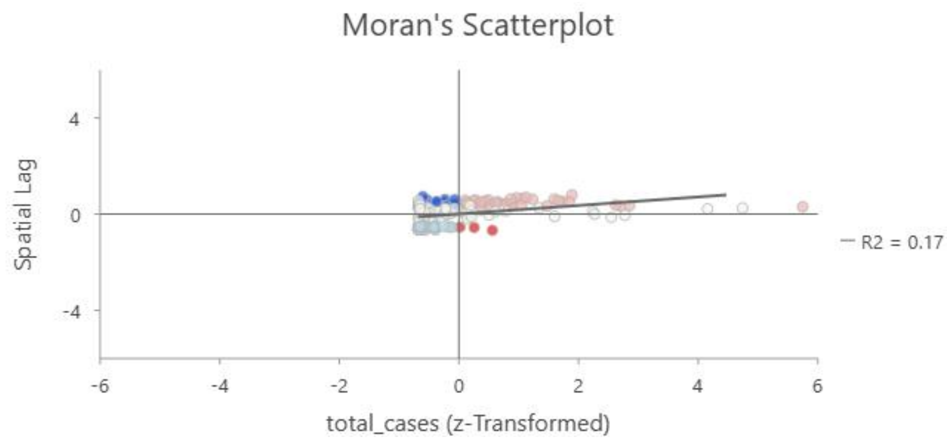
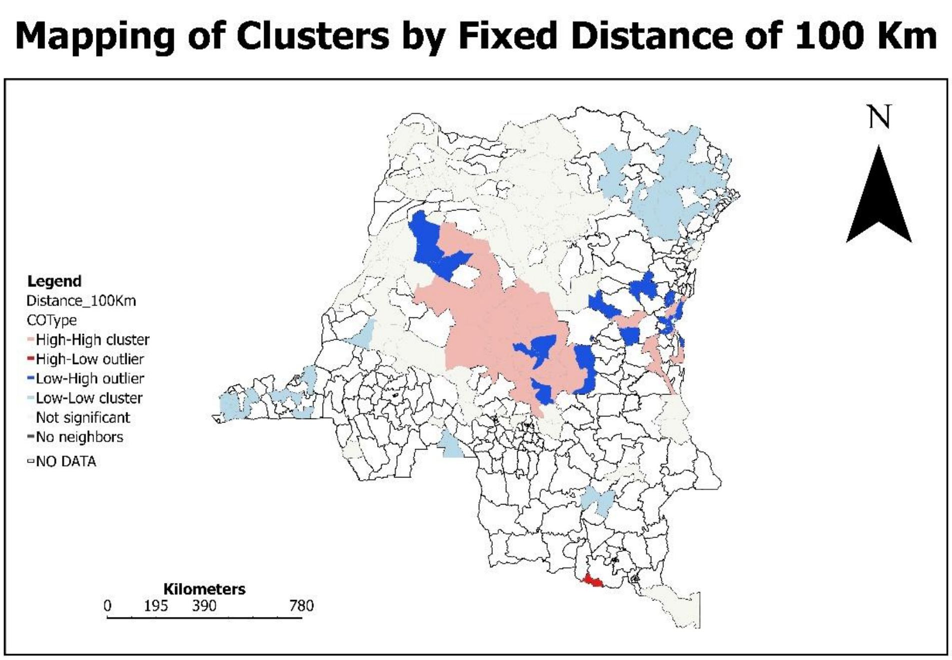


Figure 13: Result for local cluster by Fixed Distance (Default Threshold)

Fixed Distance Band (100 km)

With a more restrictive 100 km distance threshold, the Local Moran's I map identified High–High clusters in the central and extreme eastern regions of DR Congo health zones. This indicates strong localized clustering of increased Mpox cases within shorter spatial interaction ranges. Spatial outliers were detected, while Low–Low clusters in the northeastern and southern region remained consistently identifiable**.**



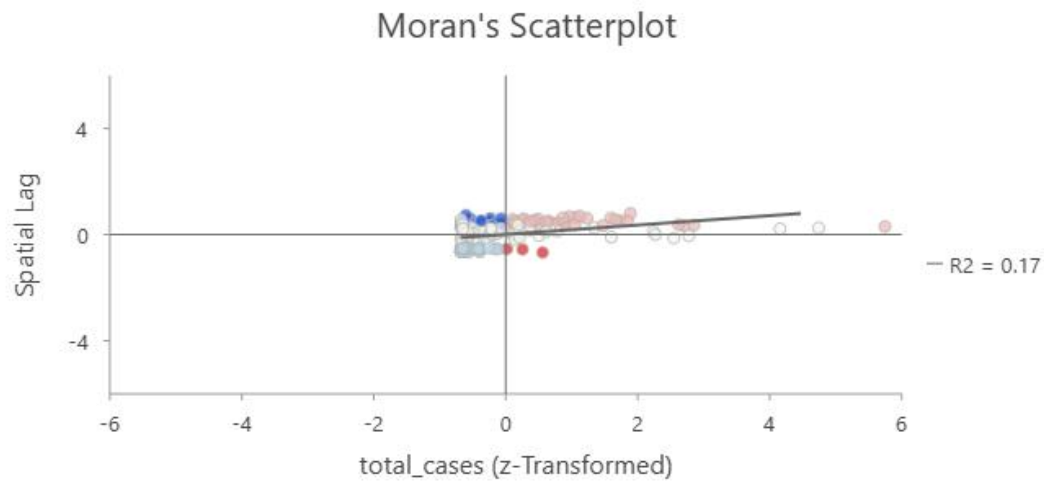
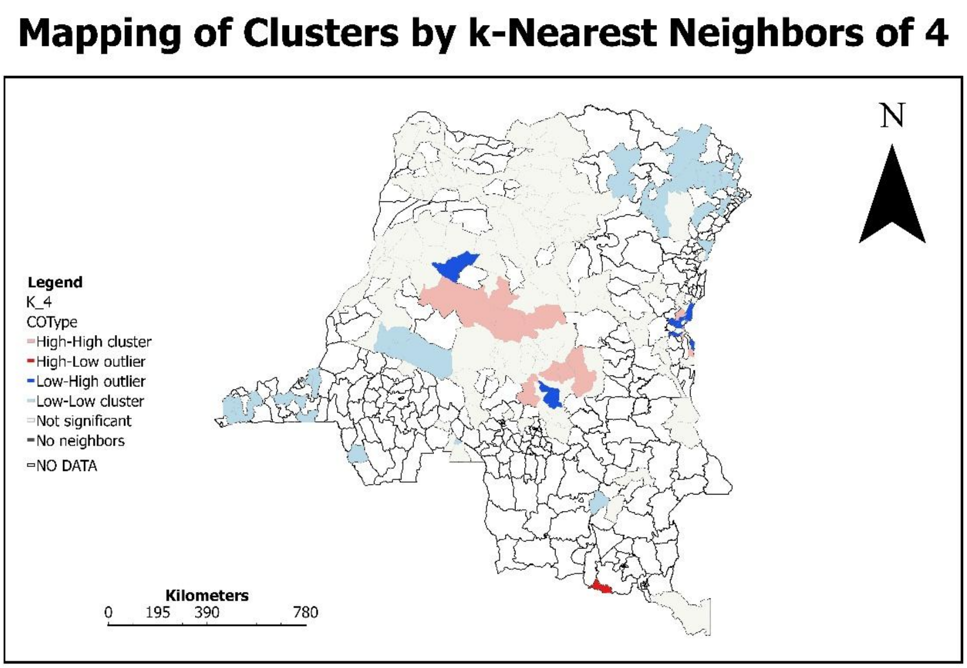


Figure 14: Result for local cluster by Fixed Distance (100 km)

k-Nearest Neighbors (4 Neighbors)

With k = 4 nearest neighbors, the Local Moran's I map identified a central High–High cluster that is less continuous than under distance-based specifications. The map detects several spatial outliers surrounding the central cluster and in eastern locations, suggesting localized inconsistencies in Mpox burden relative to nearby health zones. Northeastern and southern Low–Low clusters remain present but are more fragmented.



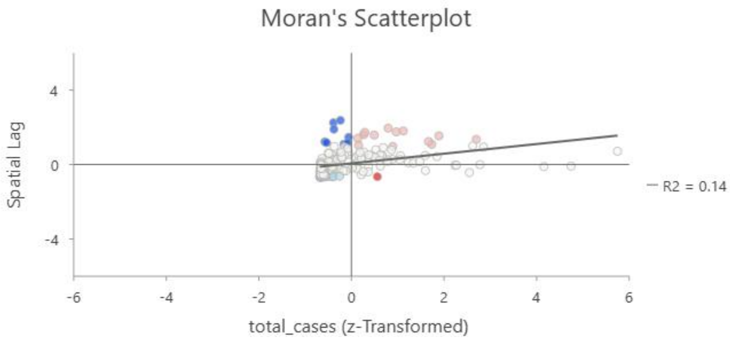
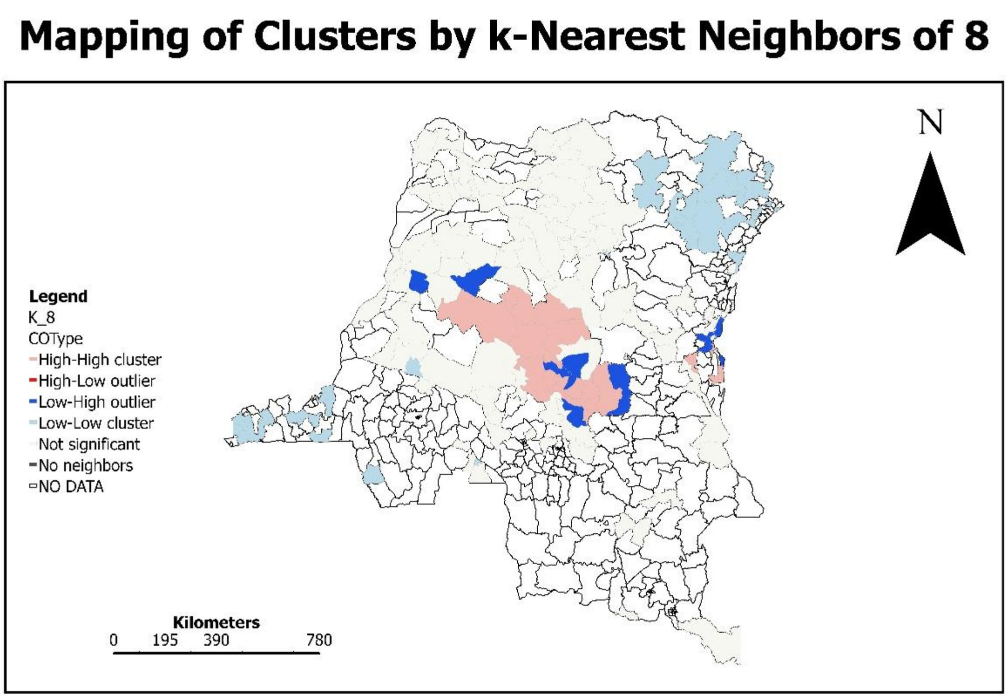


Figure 15: Result for local cluster by k-Nearest Neighbors (4 Neighbors)

k-Nearest Neighbors (8 Neighbors)

Increasing the neighborhood size to k = 8 resulted in a smoother spatial pattern, with the central High–High Mpox cluster remaining clearly identifiable but with fewer detected spatial outliers. This suggests that some localized deviations observed at k = 4 were absorbed when broader neighborhood structures were considered. Low–Low clusters in the northeastern and southern regions persisted. k = 8 detected a strong central and extreme eastern hotspots, a persistent north-eastern and southern Low-Low clusters and Low-High outliers.



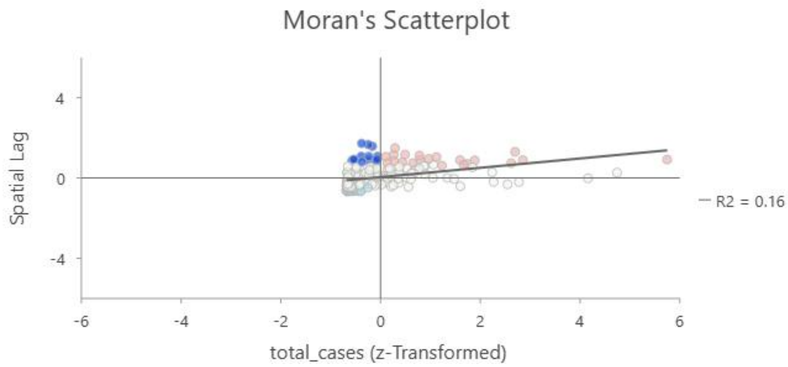


Figure 16: Result for local cluster by k-Nearest Neighbors (8 Neighbors)

5.2 Comparison of Methods Used

Comparison of Local Moran's I results across contiguity-based and distance-based, it was established that all the 5 spatial weighting methods can detect clusters and outliers in observations (dataset). All specifications identified a persistent High–High hotspots in central and extreme eastern regions of DR Congo and Low–Low clustering in the north-eastern and southern regions, indicating stable areas of high and reduced Mpox burden, respectively. Nevertheless, The Fixed Distance band both at default and 100 km threshold conceded the greatest number of hotspots in the **Central and extreme Eastern** health zones. The results also revealed that Mpox clustering in the DRC is geographically consistent but varies in spatial extent and as well as the detection of outliers.

5.3 Discussion

Multiple spatial weights were used to assess the robustness of global spatial autocorrelation results and ensure that observed clustering patterns were independent of a single neighborhood definition. All methods detected Hotspots of Mpox Risks in the **Central and extreme Eastern Region** health zones with varying geographic extent in the Democratic Republic of Congo.

To mention but a few, in the **Central Region**, health zones with persistent **High–High hotspot** of Monkey Pox in **Sankuru Province** includes Wembo Nyama, Katako Kombe, Tshudi Loto, Bena Dibele and Tshumbe. In the **Tshuapa Province**, the highly burdened Mpox risks are consistent in Monkoto, Ikela, Yalifafo, Bokungu and Bosanga. Other provinces in Central DRC with high risk of Mpox health zones are **Kasai and Maniema**

In the extreme **Eastern** health zones Mpox risks are mainly in the **Sud-Kivu Province**. These health zones include Uvira, Kitutu, Kamituga, Nundu, Shabunda and Lulingu.

I **recommend** that further research be done to know the causes of this high Mpox Risks in these provinces and their health zones.

CHAPTER 6

RESOURCE ALLOCATION AND CONCLUSION

6.1 Resource Allocation

The clustering results showed that Mpox burden is spatially concentrated, with persistent High–High hotspots in **Central** DRC and **in the extreme Eastern** health zones. These areas include the **Sankuru Province**, **Tshuapa Province and Sud-Kivu Province, Kasai and Maniema Provinces. Therefore, these provinces** should be prioritized for intensified surveillance, diagnostic capacity, and coordinated response across neighboring health zones. Spatial outliers along the margins of hotspot areas indicate transition zones that require targeted monitoring and rapid response to prevent further spread. In contrast, Low–Low clusters in the north-eastern and southern regions show consistently low burden and can be managed through routine surveillance. Overall, the findings support geographically targeted resource allocation, focusing intensive interventions in hotspot regions while maintaining baseline monitoring in low-risk areas.

6.2 Conclusion

This project aimed to evaluate the spatial concentration of weekly Mpox cases across 200 health zones in the Democratic Republic of the Congo from October to December 2025, identify areas of high risk, compare clustering patterns under different spatial conceptualizations, and derive insights to support public health resource allocation.
According to the results and the objective of this project, the following conclusions are made:

i. Mpox risk maps at health zone level under 5 different spatial weights have been produced.
ii. Spatial concentration of the Mpox risk across the health zones has been identified
iii. Comparison of clustering under contiguity-based and distance-based spatial weight has been achieved and
iv. Insights into resource allocation have been provided.

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